

Experiment 15 –POMDP

Multi-Agent Reinforcement Learning and Partially Observable Markov Decision Process (POMDP) simulation

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SOLT:B

Aim:

To implement a **Multi-Agent Reinforcement Learning (MARL) and Partially Observable Markov Decision Process (POMDP) simulation** and analyze the performance of multiple agents using reward visualization.

Procedure:

1. Create a simple partially observable environment.
2. Initialize multiple agents with independent Q-tables.
3. Provide each agent with a partial observation of the environment.
4. Select actions using an ϵ -greedy policy.
5. Calculate rewards and update the agents' Q-values.
6. Repeat the process for multiple episodes.
7. Calculate the average and final rewards of the agents.
8. Visualize the reward performance of all agents.

Python Code:

```
import numpy as np
import matplotlib.pyplot as plt
np.random.seed(1)
agents=3
states=5
actions=2
```

```

Q=np.zeros((agents,states,actions))
rewards=np.zeros((agents,50))
alpha,gamma,epsilon=0.1,0.9,0.1
for ep in range(50):
    s=np.random.randint(states)
    for a in range(agents):
        obs=max(0,min(states-1,s+np.random.choice([-1,0,1])))
        action=np.argmax(Q[a,obs]) if np.random.rand()>epsilon else
np.random.randint(actions)
        ns=max(0,min(states-1,s+(1 if action else -1)))
        r=1 if ns==states-1 else -0.1
        Q[a,obs,action]+=alpha*(r+gamma*np.max(Q[a,obs])-Q[a,obs,action])
        rewards[a,ep]=r
print("===== MARL + POMDP SUMMARY =====")
print("Agents:",agents)
print("States:",states)
print("Actions:",actions)
for a in range(agents):
    print("Agent",a+1,"Average Reward:",round(np.mean(rewards[a]),2))
    print("Agent",a+1,"Final Reward:",rewards[a,-1])
plt.figure(figsize=(8,4))
for a in range(agents):
    plt.plot(rewards[a],label="Agent "+str(a+1))
plt.xlabel("Episode")
plt.ylabel("Reward")
plt.title("Multi-Agent POMDP Performance")
plt.legend()
plt.grid()
plt.show()

```

Result:

The **Multi-Agent Reinforcement Learning and POMDP simulation** was successfully implemented. Multiple agents learned from **partial observations** and updated their Q-values based on the received rewards. The performance was evaluated using summary statistics and a comparative reward graph.

Summary Output:

```
===== MARL + POMDP SUMMARY =====  
Agents: 3  
States: 5  
Actions: 2  
Agent 1 Average Reward: 0.41  
Agent 1 Final Reward: -0.1  
Agent 2 Average Reward: 0.41  
Agent 2 Final Reward: -0.1  
Agent 3 Average Reward: 0.36  
Agent 3 Final Reward: -0.1
```

Visualization Output:

